Beyond Attackers, Midfielders, and Defenders: Identifying Positional Archetypes in the National Women's Soccer League (NWSL) through Clustering

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ABSTRACT

Positional analysis in soccer has traditionally heavily focused on the men's game. However, with the rising popularity and profitability of the women's game, exemplified by leagues such as the National Women's Soccer League (NWSL), it is increasingly important to understand how players in different positions contribute to their teams in women's soccer. We address what statistics separate midfielders from defenders and attackers, and explore the nuances within each position by identifying subgroups, such as defensive midfielders, with a focus on trends emerging between the 2023 and 2024 NWSL seasons.

To address the research gap, we use exploratory data analysis and hierarchical clustering to conduct our analyses. Our analysis suggests the existence of multiple subgroups within all positions, and reveals that the number of groups of players does not change significantly from year to year. There are multiple archetypes for each of the three main field positions, attackers, midfielders, and defenders. This approach can help teams find similar players across the league and helps viewers understand that positional subgroups exist beyond the three traditional position labels.

INTRODUCTION

The competitive nature of professional soccer leagues demands a nuanced understanding of player roles and contributions. This is no different for the National Women's Soccer League (NWSL) — the highest professional women's soccer league in the United States. Originally founded in 2013 with eight teams, the league now consists of 14 teams (as of 2025), with plans to expand to 16 teams by 2026. The league is filled with players from the United States and across the world, including some players who play internationally for their countries.

While some studies on the NWSL examine fan demographics (Meindl, 2023) and stadium locations (Chahardovali et al., 2024), there is limited research that specifically quantifies the contributions of different types of players across the league.

This paper examines whether certain statistics can effectively differentiate between defenders, midfielders, and attackers. By addressing these points, we fill a gap in the current sports analytics literature and contribute a data-driven approach to player evaluation in women's professional soccer. We specifically look to address the following questions: How can we quantify the role midfielders play offensively and defensively? Are there specific statistics that separate defenders, midfielders, or attackers?

DATA

We used publicly available data gathered by Dror and Jackson [1], accessed via the nwslR package [2], which includes season-level player statistics from 2016-2024 — overall containing data for 923 players across 8 seasons. Of the 8 seasons, only data from the 2023 and 2024 regular season and playoffs was used in this analysis; all Challenge Cup data was excluded. The 2023 and 2024 seasons were chosen to analyze because 2023 and 2024 were the latest seasons available and provided an insight into the league post-COVID.

The data was further filtered to exclude goalkeepers and only include players who played 10+ games in 2023 or 2024. There were 140 players who met this criteria in 2023 and 239 players who met this criteria in 2024. For each player, the data included 36 variables, including 4 identifiers: season, team abbreviation, player match name, games, played, and player position. Table 1 organizes all variables by whether we consider them to be offensive or defensive statistics. All variables were standardized into rate statistics, by taking the variable and dividing it by the number of games played.

Offensive Statistics	Defensive Statistics
Key Passes	Recoveries
Through Balls	Interceptions
Touches	Tackled
Crosses Open Play Successful	Tackles Won
Successful Passes Total	Tackles Lost
Total Fouls Won	Tackles Total
Passes Forward	Duels Won
Passes Backwards	Duels Lost
Loss of Possession Total	Duels Ground
Short Passes Successful	Aerial Duels Won
Short Passes Unsuccessful	Aerial Duels Lost
Shots Total	Ground Duels Won
Shots on Target	Duels Ground Lost
Long Passes Successful	
Long Passes Unsuccessful	

Table 1: Statistics separated by offense and defense

METHODS

Principal Component Analysis

To better understand which variables contributed to the most variation between players, we employed Principal Component Analysis (PCA) to find the variables that account for the most variation of the dataset. We visualized the loadings of Principal Component 1 (PC1) and Principal Component 2 (PC2), to reveal natural groupings, clusters, and outliers in the data set. PC1 and PC2 accounted for over 60% of variance in the dataset of 2023 and 2024 season data combined. They represent the two directions in the data that capture the most variation across all variables.We took the absolute sum of each variable between the two components to define the ten variables that capture the most variation in the dataset, which were: touches, successful passes, total short passes successful, loss of possession total, duels ground, passes forward, duels lost, duels ground lost, short passes unsuccessful, passes unsuccessful.

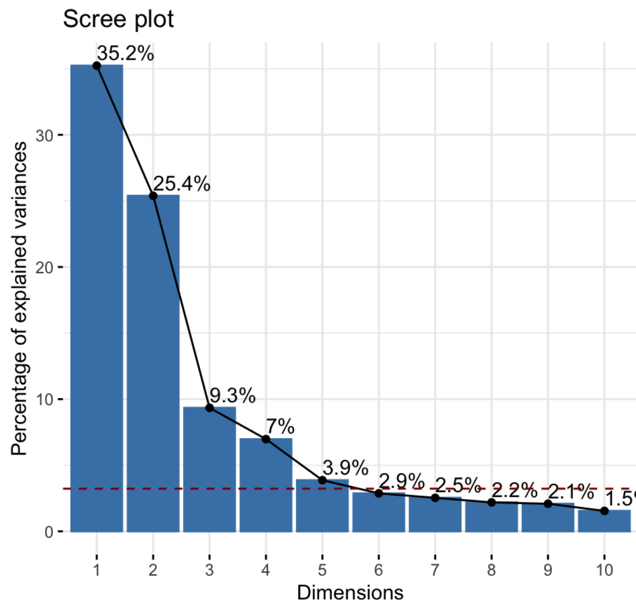


Figure 1: Percentage of explained variances across principle components in both 2023 and 2024

Hierarchical Clustering

We applied Agglomerative hierarchical clustering to group similar players together based on their statistics for the 32 offensive and defensive statistics listed in Table 1. We let N be the set of individual players, partitioned into n sets such that each player was into their own cluster. $A\ and\ B$ are subsets of N, created when partitioning N into n sets. Then using a Euclidean distance matrix, we computed the pairwise dissimilarities between each individual player; all variables were standardized on a rate scale prior to computing the Euclidean distance. To compute the dissimilarities, we used the complete linkage method, taking the maximum value of these dissimilarities $max\{d(x, y) : x\epsilon A,\ y\epsilon B|\ A, B\epsilon N\}$ [3]. After identifying the pair of clusters that are least similar, the two clusters were then fused into a new cluster. This process continued until every cluster was fused to each other. We assigned cluster labels by defining the desired number of clusters k.

Cluster Comparison

To evaluate similarities between the 2023 and 2024 clusters, we calculated the Adjusted Rand Index. The Adjusted Rand Index is a statistical measure used to evaluate the similarity between two data clustering, accounting for chance agreement. In our case, the two data clusterings were the player data sets for 2023 and 2024, only including players who played 10+ games in both seasons and were not goalkeepers. Mathematically ARI is defined as: $ARI_k(p_k^{2023}, p_k^{2024}) = \frac{N(A+D) - [(A+B)(A+C)+(C+D)(B+D)]}{N^2-[(A+B)(A+C)+(C+D)(B+D)]}$ where A, B, C, and D are counts of agreement and disagreement between clustering assignments. ARI values range from -1 to 1 where a 1 indicates agreement between the two clusters, a 0 indicates a level of agreement comparable to what you would expect from random chance, and negative values suggest less agreement than chance or strong disagreement [4].

The ARI was calculated twice, first under the criteria where number of clusters (k) equaled 7 and second when k equaled 2. We selected 7 clusters in an attempt to place players across the (non-side dependent) seven traditional field positions: striker, winger, centre-forward, holding midfielder, attacking midfielder, defensive midfielder, centerback, and fullback. We used the k = 2 as a way to differentiate more attacking players from more defensive players, and as a baseline to compare our k = 7 calculation against.

Cluster Standardization

For both seasons, we generated cluster centroids by obtaining the mean for each variable for all seven clusters. We standardized each centroid value by calculating the Z-score using the following formula: $(\textit{centroid value} - \textit{variable mean})/(\textit{variable standard deviation})$. Changes across the ten key variables were visualized using a parallel coordinate plot. A primary position was assigned for each cluster, based on which position had the highest frequency in the cluster.

RESULTS

Positional Distributions Using Principal Component Analysis

In Figure 2a, we observed that attackers and defenders were positioned on opposite sides of the scatterplot, while midfielders were distributed between them. This indicates that attackers and defenders share few similarities across the 25 variables analyzed. The trend persisted into 2024, with attackers and defenders remaining separated on the PCA plot in Figure 2b. Some midfielders were grouped with attackers, while others were grouped with defenders. The overlap between midfielders and attackers suggests the existence of attacking midfielders, whereas the overlap between midfielders and defenders indicates the presence of defensive midfielders.

Figure 2: Players colored by position across principal components 1 and 2 in 2023 (a) and 2024 (b)

The distribution of attackers, midfielders, and defenders across teams on a scatter plot of PC1 and PC2 varied throughout Figure 3a. The variation between teams indicates that no team had a unique combination of players across positions. A similar trend followed for teams in 2024, as indicated by Figure 3b.

The overall distribution of players for the entire year matched the distribution of points for each team, which indicates that if one were to take a small group of players from one section of Figure 2a or Figure 2b, none of these players would play for the same team.

Across 2023 and 2024, we also see a few defenders separated from the majority of players in Figure 2a and 2b. These players mostly all played for the North Carolina Courage (NCC), which suggests that NCC's defenders had unique statistical profiles. These NCC players played differently than their counterparts.

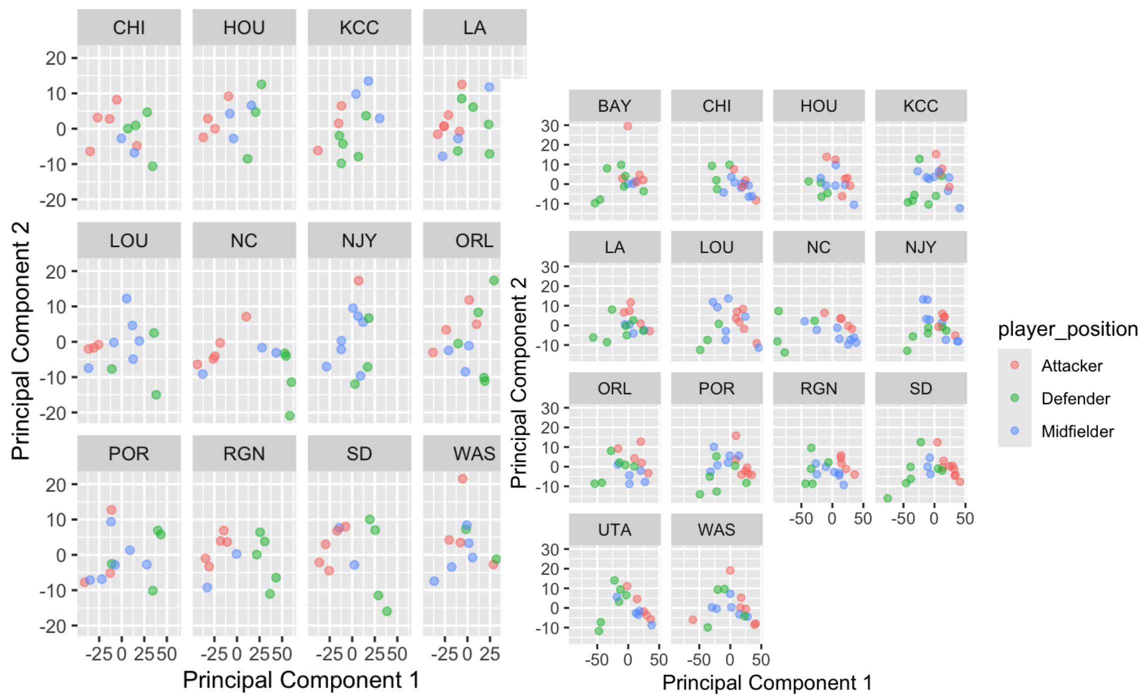


Figure 3: Players colored by position across principal components 1 and 2 faceted by team in 2023 (a) and 2024 (b)

Finding Unique Player Types by Clustering

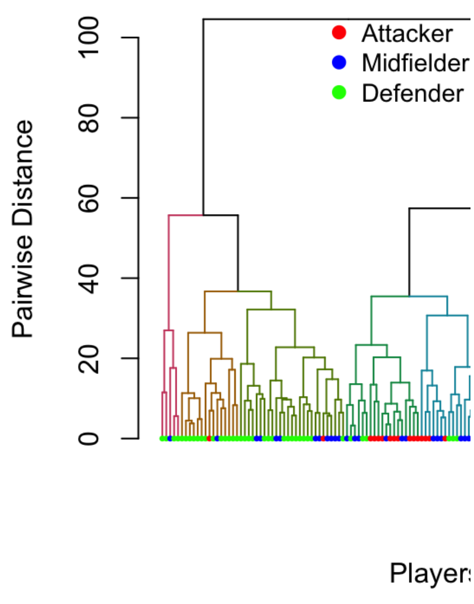
At a pairwise distance of 35, we observe 7 unique clusters in both 2023 and 2024. The 7 unique clusters demonstrate the seven unique types of players determined by the 25 different variables initially selected for analysis. The number of unique types of players stayed at 7 across 2023 and 2024 — suggesting a consistent structure in player types within the league across the two seasons. This consistency indicates that, despite any changes in the league, the fundamental ways in which players differentiate themselves based on the selected variables remained unchanged.

The dendrograms in Figure 4 (a) and (b) are a tree-like diagram used to visualize the results of

In both Figures 4 (a) and (b), there is a clear separation at the pairwise distance of 70. At this point, the players are separated into two clusters, one cluster of primarily defenders with a sprinkle of midfielders, and the other with just attackers and midfielders. The separation of defenders from attackers follows what was observed in Figure 2a and Figure 2b. In both Figure 4a and Figure

4b, there is a cluster with just four players, cluster 1 and cluster 5 in 2023 and 2024 respectively. Three players in cluster 1 in 2023 are in cluster 5 in 2024. All three of these players – K. Kurtz, R. Williams, and M. Berkeley – play for the North Carolina Courage (NCC). Inspection of their data showed that they had the highest rate of touches, averaging 30 touches more than the average player. The statistical uniqueness of Kurtz, Williams, and Berkeley suggests that NCC rely a lot on their defenders to maintain possession. The high number of touches also indicates that NCC played possession based soccer in both seasons, and prioritized building offense out of the backline — following what was understood in the "Positional Distributions Using Principal Component Analysis" section. Bay FC's (BAY) R. Kundananji was a statistical outlier among all players, even attackers.

The five player nature of cluster 5 is unsurprising, as these players: K. Kurtz, R. Williams, M. Berkeley, and N. Girma are all ball playing defenders who are asked by their teams to retain possession. As mentioned previously, Kurtz, Williams, and Berkeley played on NCC, where coach Sean Nehas asked them to play possession oriented football. Girma, of the San Diego Wave, was the central cog of San Diego's attack and defense as a rookie in 2024.



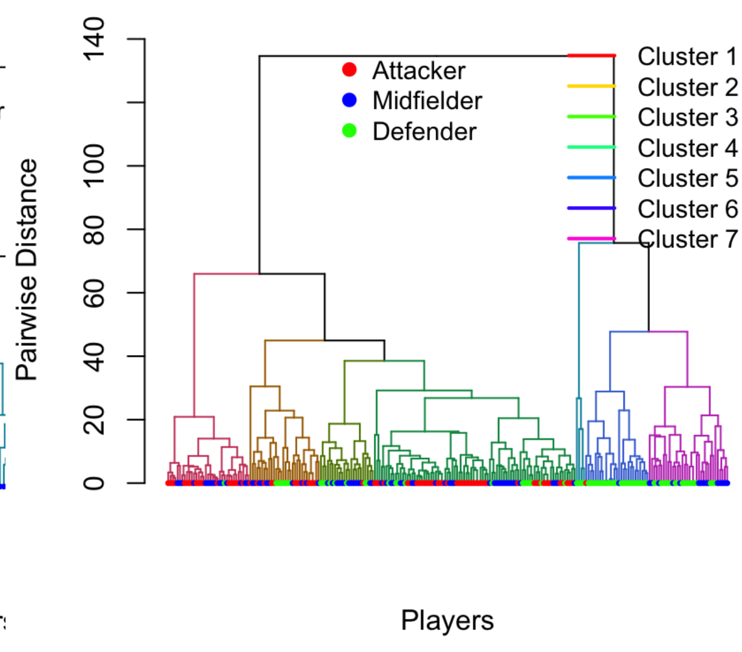


Figure 4: Dendrogram colored by player position in 2023 (a) and 2024 (b)

Team and Cluster Correlation

To examine whether player cluster assignments are independent of team affiliation, a Pearson's Chi-squared test was conducted using a frequency table of player counts per team and cluster on the data from both 2023 and 2024. The test yielded a Chi-squared statistic of 21.133 with 66 degrees of freedom and a p-value of 1 for 2023 and a Chi-squared statistic of 74.924 with 78 degrees of freedom and a p-value of 0.5777 for 2024 — indicating no statistically significant association between team and cluster membership.

By Season Player Similarity

Setting the number of clusters (k) to 7 resulted in an Adjusted Rand Index (ARI) of 0.1020024. An ARI indicates 0.1020024 weak agreement between players and their assigned cluster in 2023 and 2024, also suggesting clusters formed with k = 7 may not be meaningfully differentiating players into different types. While the league may have a similar number of unique types of players, players themselves changed the way they played between 2023 and 2024. Performance characteristics for many players were likely not consistent across the two years.

Using k = 2, we calculated an ARI of 0.3135518. This means that players' performance metrics across the two years exhibit moderate similarity between 2023 and 2024 when classified into only two types of players. For many players, performance characteristics were likely consistent across both years when grouped into these two unique player types.

Centroids Change Across Seasons

For both 2023 and 2024, there are three clusters with primarily composed attackers, and three clusters primarily composed of defenders (see Table 2 and Table 3). There is only one cluster primarily composed of midfielders (cluster 1 in 2023 and cluster 1 in 2024).

Cluster	Attackers	Midfielders	Defenders	Players Total
1	4	12	4	20
2	10	7	0	17
3	22	8	7	37
4	1	1	13	15
5	1	10	16	27
6	10	6	3	19
7	0	1	4	5

Table 2: Player Position by Cluster in 2023

According to Table 3, Cluster 2 is an attacking cluster, there are a large portion of midfielders. These midfielders can be classified as attacking midfielders, given that they are most similar to a certain type of attacker. Cluster 5 is a defending cluster, but also has a large portion of midfielders as well. These midfielders can be subclassified as defensive midfielders, given their statistical similarities to defenders. Cluster 3 is a primarily attacking cluster, but has an almost equal distribution of attackers, midfielders, and defenders; suggesting that players regardless of their position share similar statistical profiles that align more closely with attackers.

Cluster	Attackers	Midfielders	Defenders	Players Total
1	1	14	8	23
2	45	29	12	86
3	13	10	7	30
4	1	1	25	27
5	0	15	19	34
6	19	15	1	35
7	0	0	4	4

Table 3: Player Position by Cluster in 2024

According to Figure 5, defenders have a higher rate of touches, successful passes, successful short passes, and passes forward across 2023 and 2024. While players may have changed "unique player type" across season, position wise, players in 2024 played similar to players in 2023. Essentially, defenders played similarly across both seasons. Cluster 6, primarily composed of attackers and closely followed by midfielders, had a lower rate across all variables in both 2023 and 2024. The players in Cluster 6 in Figure 5b lose fewer duels, but also contribute less to their team's passing game. Across 2023 to 2024, we do not see a clear difference between how the clusters change between the various variables.

A parallel coordinates plot was chosen for Figure 5 to illustrate the change in the top ten statistics causing the most variability in the dataset across clusters. Overall, we observe that defender heavy clusters rate highly on offensive statistics like touches, successful passes, etc, while attacker heavy clusters lose possession more. The singular midfield-heavy cluster is always between the attacker and defender heavy clusters on all statistics. The change across statistics is consistent between 2023 and 2024.

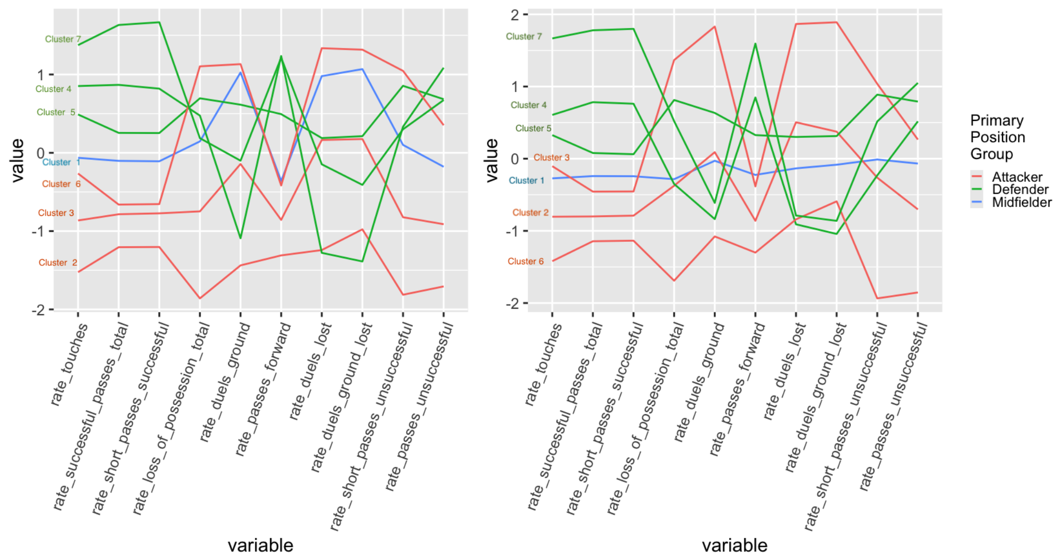


Figure 5: Cluster centroid comparison to mean by variable for 2023 (a) and 2024 (b)

DISCUSSION

Key Findings and Interpretations

Our research observed that there were seven unique groups of players in 2023 and 2024. In both seasons, there were three variations of attackers, three variations of defenders, and only variation of midfielder. We found that defenders and attackers are very statistically different, defenders contributing a higher number of touches, and successful passes (overall, shorts, and forward). Our research also found that there are many midfielders that are statistically similar to attackers (attacking midfielders), and midfielders that are statistically similar to defenders (defensive midfielders). Players generally played differently season to season, but the league's makeup of types of players and proportion of types of players did not change from 2023 to 2024.

Limitations

The league expanded from 12 to 14 seasons between 2023 and 2024, increasing the number of regular season games played from 22 to 26 and increasing the number of players from 140 to 239. Teams who made the playoffs played 1-3 more games, of which the quality of opponents increased. All these factors may have altered results in a way not yet fully understood.

Further Research

My research addresses the analysis gap in women's sports. **In my search for technical analysis of women's soccer, I found zero relevant articles.** On the men's side, Memmert (2016) [5] examined how positions differ based on running speed, while Konefal (2019) [6] analyzed the breakdown of physical and technical activities across positions. No such publicly available data or analysis exists on the women's side. My research on positioning in the NWSL across seasons helps increase understanding of how individual players contribute to positionality in their teams and the league. I have identified and quantified the statistics that differentiate players, providing clarity on the statistical boundaries between defenders, midfielders, and attackers. Furthermore, it redefines positional archetypes within the NWSL — showing that defensive midfielders and attacking midfielders are statistically quantifiable, just like they are on the pitch. Similarly, the research I've conducted shows that a player's role can (and likely will) vary year to year.

This data can also identify similar players across the league. If a player is injured for the season, teams can use this information to find and trade for another player who has historically played similarly within the league.

Similar analysis in other top level women's professional soccer leagues such as the Barclays Women's Super League, Liga MX Femenil, Damallsvenskan, Frauen Bundesliga, or A League could also be insightful. It would certainly be interesting to see if positionality based on the same criteria changes across leagues, or even continents.

An expansion of this project using player tracking data would be helpful to better understand where the different clusters of players hang out across the field.

Conclusion

In conclusion, we used cluster analysis to model distinctions between player types and how players fit in the league and on their teams within the 2023 and 2024 NWSL seasons. While the league witnessed an expansion and increased game frequency, the density of player types remained consistent, underscoring that play stayed consistent despite major changes to the structure of the league. This study fills a critical gap in research on play in women's sports and introduces a more comprehensive understanding of positionality and player similarities. Future research can expand the understanding gained in this research to different leagues abroad.

REPRODUCIBILITY STATEMENT

The code and data in this analysis is reproducible and available on Github at: https://github.com/tp56/nwsl-archetypes

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